Physics-0   2026 Spring

Homework 8

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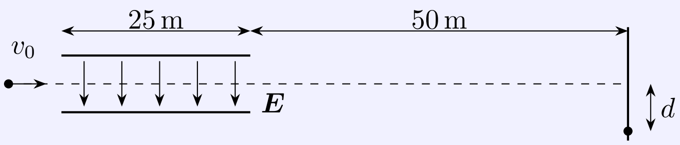
*All of my homework is available at `https://github.com/lacampanella233/Homework.git`.

Problem 1. *Consider a particle of mass m, charge q, and initial speed $v_0 = 5\times 10^3\,\mathrm{m/s}$ projected into a uniform electric field between two parallel plates of length $25\,\mathrm{m}$. The electric field is directed downward between the plates with magnitude $|\boldsymbol{E}| = 800\,\mathrm{N/C}$ and vanishes outside of the plates. After passing through the field, the object reaches a wall at a distance of $50\,\mathrm{m}$, where it is found to have been deflected downward a distance $d = 1.25\,\mathrm{m}$. Ignoring gravity and air resistance, compute the object's charge-to-mass ratio q/m.*



Solution. Denote the length of the parallel plate by $l = 25\,\mathrm{m}$ and the distance between the wall and the plate by $L = 50\,\mathrm{m}$.

The particle will stay at the uniform electric feild for a time $t_1 = L/v_0$ and then move in the free space for a time $t_2 = l/v_0$.

The acceleration of the particle in the electric field is $a = qE/m$. Then the virtical speed of the particle when it leaves the electric field is $v_y = at_1$. Hence the virtical displacement of the particle in the electric feild is

$$d_1 = \frac{1}{2}at_1^2,$$

and the virtical displacement of the particle in the free space is

$$d_2 = t_2 v_y.$$

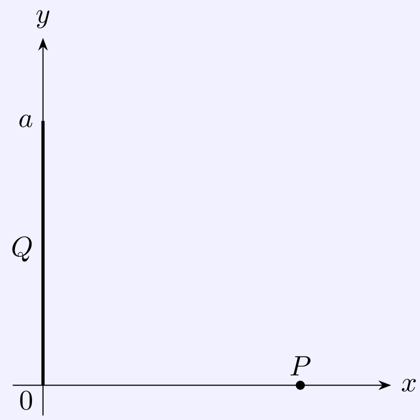
By the relation $d = d_1 + d_2$, we can solve the charge-to-mass ratio of the particle as

$$\frac{q}{m} = \frac{2v_0^2 d}{El(l+2L)} = 25\,\mathrm{C/kg}.$$

Problem 2. *Positive charge Q is distributed uniformly along the positive y-axis (between $y = 0$ and $y = a$) as shown in the figure below.*

(a) Calculate the x- and y-components of the electric field produced by the charge distribution Q at the point P on the positive x-axis.

(b) Compute the leading contribution to the magnitude of the electric field in the limit $x \gg a$. Does the result fit your expectations? Explain why.



Solution. (a) Denote $P = (x_0, 0)$. The x- component of the electric feild is given by

$$E_x = \int_{y=0}^{a} \frac{Q\,\mathrm{d}y}{a} \frac{1}{4\pi\varepsilon_0} \frac{x_0}{(x_0^2+y^2)^{3/2}} = \int_{\theta=0}^{\arctan(a/x_0)} \frac{Qx_0\,\mathrm{d}\theta}{a\cos^2\theta} \frac{1}{4\pi\varepsilon_0} \frac{\cos^3\theta}{x_0^2} = \frac{Q}{4\pi\varepsilon_0} \frac{1}{x_0\sqrt{x_0^2+a^2}}.$$

The y- component of the electric feild is given by

$$E_y = -\int_{y=0}^{a} \frac{Q\,\mathrm{d}y}{a} \frac{1}{4\pi\varepsilon_0} \frac{y}{(x_0^2+y^2)^{3/2}} = -\int_{\theta=0}^{\arctan(a/x_0)} \frac{Qx_0\,\mathrm{d}\theta}{a\cos^2\theta} \frac{1}{4\pi\varepsilon_0} \frac{x_0\tan\theta\cos^3\theta}{x_0^3}$$
$$= -\frac{Q}{4\pi\varepsilon_0 a x_0}\left(1 - \frac{x_0}{\sqrt{x_0^2+a^2}}\right).$$

(b) The leading contribution to the magnitude of the electric feild in the limit $x_0 \gg a$ is given by

$$E = \sqrt{E_x^2 + E_y^2} \approx \frac{Q}{4\pi\varepsilon_0 x_0^2}.$$

This result fit the expections, since the system can be treated as a point charge Q when $x_0 \gg a$.

Problem 3. *In class (and the problems above), we have considered the electric field of different charge configurations in three spatial dimensions, where the electric field of a single point particle in spherical coordinates reads*

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{kq}{r^2}\,\hat{\boldsymbol{r}}.$$

In this problem, we will consider a world that has only two spatial dimensions. The electric field of a point particle in two dimensions is given by

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{\tilde{k}q}{r}\,\hat{\boldsymbol{r}},$$

where the particle is located at the origin of a polar coordinate system (r, θ) and the constant $\tilde{k}$ has units of $\mathrm{N\,m/C^2}$.

(a) Consider a test particle of charge q_0 at a location $\boldsymbol{r}$. What is the potential energy of the system? How much work does it take to move the particle to infinity?

(b) Compute the electric field of an infinite long line with uniform linear charge density λ. How does this result compare to the electric field of an infinite plane in three dimensions?

Solution. (a) The potential is

$$U = \int_{x=r_0}^{r} \frac{\tilde{k}qq_0}{x}\,\mathrm{d}x = \tilde{k}qq_0 \ln\left(\frac{r_0}{r}\right) = -\tilde{k}qq_0 \ln\left(\frac{r}{r_0}\right),$$

where we define the potential energy is 0 when $r = r_0$. Hence the work to move the particle to infinity is also infinity.

(b) Suppose the line charge is along the y-axis. The electric feild at a point $P(x_0, 0)$ on the x-axis is given by

$$E_x = \int_{-\infty}^{+\infty} \frac{k\lambda\,\mathrm{d}y x_0}{(x_0^2 + y^2)^{3/2}} = \int_{-\pi/2}^{\pi/2} \frac{k\lambda x_0^2 \cos^3\theta\,\mathrm{d}\tan\theta}{x_0^3} = \frac{2k\lambda}{x_0}.$$

This result is similar to the electric feild of a point charge in two dimensions, which is also proportional to $1/x_0$.